

Grid connected electricity generation from renewable sources: Uzundere I 63.0 MW Hydroelectric Power Plant Project, Turkey



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GAIA CARBON FINANCE

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1 PROJECT DETAILS

1.1 Summary Description of Project

Karadeniz Elektrik Üretim Sanayi A.Ş. (Karadeniz, hereafter) has installed a large-scale runoff-river hydroelectric power plant (HEPP) with an installed capacity of 63.068 MWm / 62.152 MWe on the Uzundere creek within the province of Rize in Turkey. The objective of the project activity is to generate electricity and supply it into the public grid. Uzundere I HEPP was validated as a Voluntary Emission Reduction (VER) project. Due to its significant contribution in diminishing carbon emissions and protecting the climate as well as due to significant additionality issues discussed in the PDD, this project fulfilled the VCS standard and qualifies for carbon finance. In addition, the project highly supports the sustainable economic development in the region.

In the baseline scenario of the project, the project will have effect that power production of existing generators, attached to the grid, will be reduced the electricity generation from these existing generators. Since Turkey's grid mainly consists of thermal power plants, this would have resulted in GHG emissions. However, in the project scenario, the project will generate electricity from hydropower and will result in emission reductions in parallel with its electricity generation figures. Implementation of the project will consist of construction of the following main items:

- Two diversion weirs, where water from the river is diverted into conveyance pipes;
- Total length of the conveyance tunnel is 11,324 m.
- Power house with Pelton type vertical turbines;

In the power house, two vertical shaft Pelton turbines will be installed, each with 31.534 MWm / 31.076 MWe (total capacity 63.068 MWm / 62.152 MWe). There are two generators attached to the facility. Generators have power factor of 0,9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

Starting date of the project activity is May 27th, 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine has been November 7th, 2010. The carbon crediting period and therefore the monitoring starts when the plant commences electricity generation through the first turbine which started to supply electricity to the grid on May 27th, 2010. The crediting life is a maximum of ten years which may be renewed at most two times. The first feasibility study for the project was designed in June 2005. After that, the detailed project design work was completed and search for financing of the project started in September 2006. In 2006, the license was granted for that first feasibility study. Early 2007, the first civil works were started for the project. The same year, it was decided to increase the installed power to 63 MW by the board of KARADENİZ. Early 2008, the application for that increased installed power was made with a new feasibility study. After the more detailed studies, the installed capacity was decided as 62.152 MWe and revised license was granted at late 2009.

The first turbine started to operate on 27th of October 2010 and the second one started operation on 7th of November 2010, and the HEPP is operational since then.

This first monitoring period covers the dates between 27th May 2010 and 31st December 2020 (both days inclusive). Total Emission reductions achieved in this period is 85,126.69 tonnes CO₂e

The theoretical annual average energy production of the facility is based on the waterflow data which was collected by “General Directorate of State Hydraulic Works” of Turkey which is the primary executive state agency of Turkey for national overall water resources planning, managing, execution and operation. The annual average electricity production figure is based on the theoretical annual production figures with historical annual water flows.

1.2 Sectoral Scope and Project Type

The subject project is not a grouped project. According to domestic regulations, with an installed capacity of 63.068 MWm / 62.152 MWe, the Uzundere I HEPP is qualified as a large-scale project.

According to Annex A of the Kyoto Protocol, the Project fits in:

Sectoral Scope Number : 1

Sectoral Scope : Energy Industries -Renewable Energy

1.3 Project Proponent

Name	Karadeniz Elektrik Üretim San. A.Ş.
Organizational category	Private Company
Address	Fahrettin Kerim Gökyay Cad. No:36 Altunizade 34662 Uskudar ISTANBUL-TURKEY
Main activities	Karadeniz Elektrik Üretim San. A.Ş. is a subsidiary of Eksim Group which is mainly involved with in renewable energy, real estate & construction, mining, food and commodity trading such as grain (wheat, soy, corn etc.) in Turkish domestic market and international market.
Contact	Mr. İsmail Yasin Erkan +90 216 544 24 00 ismail.erkani@ilteketisim.com Mr. İsmail Yasin Erkan is the project coordinator of the Uzundere I Project, and responsible for the overseeing the carbon asset development activities on KARADENİZ side coordinating data flow to carbon asset development company (Gaia Carbon Finance), conducting relationships with the third parties such DOE, potential buyers etc.)

Karadeniz Elektrik Üretim San. A.Ş whose contact details given above is the project participant.
Other Entities Involved in the Project

There are no other entities involved in the project.

1.4 Project Start Date

The start date of the project activity is 27th May 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine is 7th November 2010.

1.5 Project Crediting Period

The crediting period of the project begins with the commissioning of the plant. The carbon crediting period and therefore the monitoring starts when the plant commences electricity generation. The crediting life is a maximum of ten years which may be renewed at most two times. With respect to the project activity, the project's first crediting period is from 27th May 2010 to 31st December 2020.

1.6 Project Location

The project is located in the Black Sea Region of Turkey, within the province of Rize. The project activity is located on the Uzundere creek. Prior to project initiation, the project area hosted some forestland and agricultural activities in small-scale lands (the most common crop in the region is tea). Because the land structure is broken and partite, agricultural economics has not developed in the region. Also, livestock production is not one of the main livelihoods in the region anymore, because it lost its value in time in parallel with the decrease in transhumance activities. The altitude at the powerhouse location is 817.59 m. The project site had been vacant as the project is a new power unit and it did not replace an old plant. The project site was treasury land. The geographical locations of Uzundere I HEPP are 40° 50' 16'' - 40° 54' 04'' North and 40° 45' 4'' - 40° 49' 42'' East, respectively.

1.7 Title and Reference of Methodology

Since the project is an installation of a new grid-connected renewable power plant, the baseline scenario is formulated in Approved consolidated baseline and monitoring methodology ACM0002, Version 11¹, titled as "Consolidated baseline methodology for grid-connected

¹

http://cdm.unfccc.int/filestorage/H/G/Y/HGY3TLRFPQVM016WA4I7XCZD92KE5S/EB52_repan07_ACM0002_ver11.pdf?t=NU98bTd6NHE2fDD_OxCAW7I_KAUyoOp6hNEK

electricity generation from renewable sources". In addition, "Tool to calculate the emission factor for an electricity system, Version 02"² has been used.

2 IMPLEMENTATION STATUS

2.1 Implementation Status of the Project Activity

Early 2007, the first civil works were started for the project and the plant was commissioned on 27th May 2010, when one of the two vertical shaft Pelton turbines in the plant commences electricity generation. The starting date of the project activity for the other turbine is 7th November 2010. The project activity has been connected to the Turkish interconnected electricity grid; therefore the produced electricity has been transmitted to TEIAS as per the agreement signed with Karadeniz Elektrik Uretim Sanayi A.S. During the first monitoring period (27/05/2010 to 31/12/2011), no emergencies or abnormal situations have been experienced.

Power plant operated normally during this monitoring period. There have been no emergencies happened to the monitoring system in this monitoring period. No critical equipment exchange that was performed during the monitoring period. No faulty measurement or no measurement differences between the meters occurred. Therefore, no calibration happened. The electricity generation of the power plant also realized close to the estimation in the PDD.

2.2 Deviations from the Monitoring Plan

During the monitoring plan, the following deviations are determined and the monitoring plan is revised:

- The source of the parameter "Quantity of net electricity generation supplied by the project plant/unit to the grid in year y" was given as "Monthly Meter Reading Protocols" in the VCS PD. However, since TEIAS started to measure the produced electricity remotely, Monthly Meter Reading Protocols are not available. Therefore, the source of the parameter is defined as "Market Financial Settlement Center (PMUM) records which includes the exact electricity productions and tractions. PMUM records are the basis of the invoices made out to TEIAS.
- The source for the cross-check of the amount of produced electricity is also revised as "OSF Forms" which are prepared by TEIAS after the remote measurements of the meters and sent to the plant engineer by e-mail at the end of each month.
- Monitoring way for A_{PJ} was defined as "The readings will be done during the period when the water flow is high to have the largest reservoir area." However, the monitoring plan is revised as "Based on the technical drawing, the reservoir area is calculated by a computer program", since it is difficult to determine the maximum water level for the monitoring period and calculate the reservoir area by obtaining GPS readings.

2.3 Grouped Project

The project activity is not a grouped project.

² <http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v2.pdf>

3 DATA AND PARAMETERS

3.1 Data and Parameters Available at Validation

Data Unit / Parameter:	EF _{grid,CMY}
Data unit:	tCO ₂ e/MWh
Description:	Baseline emission factor: the combined emission factor of the project grid system.” Tool to calculate the emission factor for an electricity system”
Source of data:	Source from the registered PDD for the Project.
Value applied:	0.559
Purpose of the data:	Used for baseline emission calculation.
Any comment:	N/A

3.2 Data and Parameters Monitored

Data Unit / Parameter:	EG _{Facility,y}		
Data unit:	MWh		
Description:	Quantity of net electricity delivered to the grid in year y		
Source of data:	Market financial Settlement Center (PMUM) records		
Description of measurement methods and procedures to be applied:	Two sets of meters have been used continuously for measurement. Then the net electricity generation supplied by the project to grid has been calculated.		
Frequency of monitoring/recording:	Continuous monitoring and monthly recording		
Value monitored:	2010 figure is 33,826.52 MWh and the 2011 figure is 118,294,09 MWh		
Monitoring equipment:	Relevant information regarding the main meter and the control meter have been provided as follows:		
	Meter	Main Meter	Control Meter
	Brand	ACTARIS	ACTARIS
	Type	SL761B071	SL761B071
	Serial Number	53064430	53064429

	The meters are firsthand and the supplier company commits that these meters fully conforms or exceeds all relevant IEC standards including those dealing with electronic metering equipment IEC61036 for class 1 equipment and IEC60687 for class 0.2S equipment. There is no possibility for human error in the measurement of the electricity. All the measurements and calculations are done via tested meters. The calibration of the electricity meters are the responsibility of TEIAS and the project owner has no control over this calibration process. The meters were sealed by TEIAS personnel. The necessary maintenance and calibration are the responsibility of TEIAS.
QA/QC procedures to be applied:	As per "Tool calculate the emission factor for an electricity system" and also cross-check measurement with on-site records.
Calculation method:	$EG_{\text{Facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$
Any comment:	-

Data Unit / Parameter:	Cap _{PJ}
Data unit:	MW
Description:	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data:	Project site
Description of measurement methods and procedures to be applied:	Supplier information on the equipment
Frequency of monitoring/recording:	Yearly
Value monitored:	62.152 MWe (63.068 MWm)
Monitoring equipment:	The data is monitored from the electricity generation license which was granted by Energy Market Regulatory Authority. The most up-to-date generation license has been provided. The installed capacity figure is 31.534 MWm / 31.076 MWe for each of the two turbines. The total installed capacity of the project is 63.068 MWm / 62.152 MWe.
QA/QC procedures to be applied:	Supplier information on the equipment and

	existence of the equipment will be checked
Calculation method:	N/A
Any comment:	-

Data Unit / Parameter:	A _{PJ}
Data unit:	m ²
Description:	Surface of the regulation pond after the implementation of the project activity, when the reservoir is full
Source of data:	Project site
Description of measurement methods and procedures to be applied:	The area was calculated by making coordinate readings around the border of the reservoir through topographical surveys or GPS readings. The technical drawing of the regulator area is based on topographical surveys. Based on the technical drawing, the reservoir area is calculated by a computer program.
Frequency of monitoring/recording:	Yearly
Value monitored:	7,275.54 (The company record on the measurement of the reservoir area can be found in Appendix I)
Monitoring equipment:	N/A
QA/QC procedures to be applied:	The technical drawing of the regulator area obtained from final design of the project is computerized and the potential lake area for the flood discharge (possible maximum water level) is calculated.
Calculation method:	N/A
Any comment:	-

3.3 Description of the Monitoring Plan

All relevant baseline emission factors are defined ex-ante. Hence, there are three sets of data to be monitored during the project activity.

The first one is the amount of electricity fed into the grid. There is already an electricity meter installation plan to provide the data that is required by the monthly invoicing process to TEIAS. Uzundere I HEPP is a new power unit which did not replace a fossil fuel plant on the project site. Karadeniz is responsible for providing the PMUM records which shows the monthly electricity generation of the company. In addition to the metering system, generation of Uzundere HEPP I can be crosschecked from on-site measurement records. The PMUM records can provide both the net electricity generation less transmission loss and electricity generation deducted transmission loss.

The purpose of the monitoring plan is to ensure that the monitoring and calculation of emission reductions of the proposed Project within the crediting period is complete, consistent, clear and accurate. The project is operated by Karadeniz which ensures the overall site management in accordance with Turkish Laws and technology providers' guidelines. The major quality assurance of the monitoring relies on the reporting system that has been regularly checked by the Board of Karadeniz Elektrik Uretim Sanayi A.S.

The monitoring has been performed in-house by the project proponent:

1. Plant Engineer is responsible for the control of the electricity supplied to the grid and imported from the grid with TEIAS. The electricity measurements are made by TEIAS remotely and the plant engineer is informed by TEIAS by e-mail regarding the electricity measurements by an e-mail each month. (An excel sheet called "OSF Form – Dengeleme Birimi Sayaç Değerleri Formu" which includes the hourly electricity generations and imports is sent to the plant engineer from TEIAS.) The plant engineer checks these electricity measurement records and reports to the Operations Manager of the plant.
2. Mechanical Engineers ensure that all the instrumentations and devices to perform the monitoring are working properly.
3. Accounting Manager is in charge of providing the electricity sales receipts to the Operations Manager of the plant.
4. Operations Manager is the VER coordinator. He is in charge of:
 - a. Ensuring that instrumentations and devices are available and properly suited to perform efficiently the monitoring.
 - b. Communicating and coordinating the monitoring tasks of all business units.
 - c. Developing, executing, analyzing and improving the VER Monitoring/Reporting Procedures. This includes the crosschecking and consolidation (with multiple sources whenever possible) of the data obtained from the electrical engineers and the accounting manager. He also recorded this operation properly to be able to provide it to the DOE during the verification process.
 - d. Calculating and report the emission reductions,

Please refer to Figure 1 where the site organizational chart is presented.

In order to verify the generated units of emission reductions, the VER coordinator, Operation Manager has prepared an annual Report of Vigilance of the Project, in which the following important aspects include:

- Year
- Net electricity supplied by the project activity to the grid (in MWh)
- Annual gross electricity generation(in MWh)
- Annual electricity consumption (in MWh)

- Calculation of the emissions reductions: ERs per year (in tCO₂e/yr) produced from the activity of the project

Each year the monitoring report will submit to DOE for the verification. The report covers the monitoring of grid-connected power generation, check report; report on calculation of the emission reductions and records of monitoring instrument repair and calibration, etc.

Data will be recorded for each crediting period and maintained at least 2 years after its end. The company will establish a dedicated maintenance system to ensure the data availability for the required period.

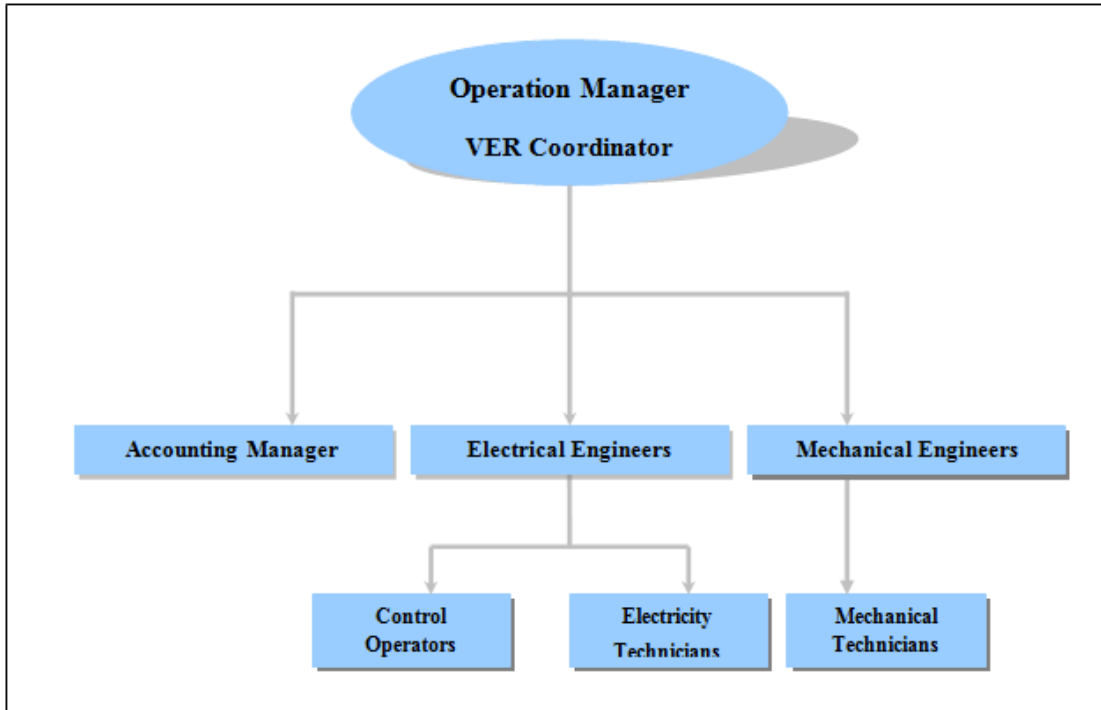


Figure 1: Site Organizational Chart

In the powerhouse, there will be two electricity meters in order to measure the produced electricity accurately. One is the main meter for measuring and the other is the check meter for control. Both of these meters are metering the energy in two directions (consumption and production). If there is a measuring difference between these two meters and one of the parties (TEIAS or the company) requests for calibration of the meters, in this case, the meters will be calibrated without waiting for the periodic calibration date. (TEIAS System Usage Agreement, Art 3, B./2./b))³ This calibration process is done by another third party under the control of TEIAS. The company is not responsible for calibration of the meters in Turkey according to the local standards.

³ www.teias.gov.tr/sistemkullanim1.doc

The electricity meters, installed to the plant are owned by Actaris has calibrated the meters in the year 2009. The model of the meters is Actaris SL761B071 and the serial numbers of the main and check meters are 53064430 and 53064429, respectively.

The meters are firsthand and the supplier company commits that these meters fully conforms or exceeds all relevant IEC standards including those dealing with electronic metering equipment IEC61036 for class 1 equipment and IEC60687 for class 0.5S equipment.⁴ In addition, the company made a test on 01 June 2010 to verify if the meters measure the electricity production accurately. The results show that the meters measure the electricity with a 0.12% error which is in line with the expected range. There is no possibility for human error in the measurement of the electricity. All the measurements and calculations are done via tested meters. If there is a measuring difference between these two meters and one of the parties (TEIAS or the company) requests for calibration of the meters, in this case, the meters will be calibrated without waiting for the periodic calibration date. (TEIAS System Usage Agreement, Art 3, B./2./b))⁵ This calibration process is done by another third party under the control of TEIAS. The company is not responsible for calibration of the meters in Turkey according to the local standards.

The Electrical Engineers will receive sufficient and continuous training in terms of monitoring and verification on aspects such as meter's reading and calibration and reading's recording, adjustment and reporting every year. If new personnel are hired, they will have to follow up a training program and will be trained in the specific skills required to carry out the Monitoring Plan.

The monitoring of the power generation and recordings are done under the supervision of the plant manager. Every month, the measurement of electricity is made by TEIAS remotely and the plant engineer is informed by TEIAS by e-mail regarding the electricity measurements by an e-mail each month. *(An excel sheet called "OSF Form – Dengeleme Birimi Sayaç Değerleri Formu" which includes the hourly electricity generations and imports is sent to the plant engineer from TEIAS. If there is a network problem and it is not possible to measure the electricity remotely, a representative from TEIAS visits the project site and measures the electricity by connecting to the meters directly. However, since there was not a network problem during the Monitoring period, all measurements were made remotely).* In addition, all production figures which are subject to sales to the grid and are agreed with PMUM (Market Financial Reconciliation Centre), can be accesses from PMUM's web site by the seller.

Therefore, net electricity production figures⁶ announced by PMUM will be used in emission calculation figures. An excel file with the monthly and annual generation is attached to this monitoring report. The power generated and supplied is monitored continuously and is recorded on a daily basis and aggregated into monthly totals.

The second data to be monitored is the installed capacity of the project activity. The installed capacity only changes if the project design is changed. If there is any changes in the installed capacity of the project, EMRA has to be informed about this change. The company will report the installed capacity changes in each monitoring period, if there is any and this information will

⁴ <http://www.actaris.com/html/index.php?module=actaris&func=viewmedia&idmp=9468>

⁵ www.teias.gov.tr/sistemkullanim1.doc

⁶ Net electricity production figure = Electricity generation (operation base) – Electricity traction from grid

be checked from EMRA website and the current license of the project. The plant manager will be responsible to provide the data. The installed capacity of the project is 62.152 MWe.

The third data to be monitored is the reservoir area. This data will be calculated from coordinate readings which are gathered through topographical surveys or GPS readings. The data will be collected by the project owner during the period when the water flow is high. However, since determining the maximum water level is difficult, to be on the conservative side, this parameter is measured from technical drawings of the project. The technical drawing of the regulator area obtained from final design of the project is computerized and the potential lake area for the flood discharge (possible maximum water level) is projected. The lake area is calculated by this computer program as 7,275.54 m².

The data will be collected and stored by Karedeniz not only during the crediting period but also two years after the last issuance of VCUs to Uzundere I HEPP project for that crediting period.

The emission factors are calculated ex-ante for the period of ten years. The combined margin will be recalculated through ACM0002 / Version11, any time the crediting period is renewed. Please note that the project is a zero emission generating activity and no particularly sensitive or critical sustainable development indicator was identified.

Data	Uncertainty Level of Data (High/Medium/Low)	Explain QA/QC Procedures Planned for this Data or why such procedures are not necessary
Quantity of net electricity delivered to the grid	Low	The metering equipment will be properly calibrated and checked periodically for accuracy, to ensure that any error resulting from such equipment shall not exceed +0.2% of full scale rating. Rules for meter accuracy are stated in the TEIAS connection agreement. In addition to the main meter, a redundant meter will be installed to exclude the possibility of data loss.
Installed capacity of the project after the implementation	Low	The installed capacity of the project is equal to the total installed capacities of the equipment implemented at the beginning of the project's life. The data is monitored from the electricity generation license which was granted by Energy Market Regulatory Authority. The most up-to-date generation license is provided for each Monitoring period. Also, supplier information on the equipment and existence of the equipment is checked.
Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full	Low	The area was calculated by making coordinate readings around the border of the reservoir through topographical surveys or GPS readings. The technical drawing of the regulator area is based on topographical surveys. Based on the technical drawing, the reservoir area is calculated by a computer program. The technical drawing of the regulator area obtained from final design of the project is computerized and the potential lake area for the flood discharge (possible maximum water level) is calculated.

4 TABLE 1: DATA TO BE MONITORED AND THE MONITORING PROCESSQUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Accordingly the baseline emissions BE_y are calculated as following:

$$BE_y = EG_{PJ,y} \times EF_{grid, CM, y}$$

Where:

BE_y	Baseline emissions in year y (tCO ₂)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)
$EF_{grid, CM, y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “ <i>Tool to calculate the emission factor for an electricity system</i> ”
Y	Refers to a given year

Since the project is an installation of a new grid-connected renewable power plant, the baseline scenario is formulated in ACM0002, Version11: “*Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the Combined Margin (CM) calculations described below*”.

The baseline emission factor is the weighted average of the Operating Margin Emission Factor and Build Margin Emission Factor as per ACM0002 Version 11 guideline recommends equal weight values for hydropower projects as seen in the formula below:

$$EF_{grid, CM, y} = EF_{grid, OM, y} \times w_{OM} + EF_{grid, BM, y} \times w_{BM}$$

Where:

$EF_{grid, BM, y}$	=	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EF_{grid, OM, y}$	=	Operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
w_{OM}	=	Weighting of operating margin emissions factor (%)
w_{BM}	=	Weighting of build margin emissions factor (%)

The default values recommended by “Methodological tool (Version 01) Tool to calculate the emission factor for an electricity system” for w_{OM} and w_{BM} for activities other than wind and solar power generation projects are 0.5 and 0.5, respectively.

Therefore resulting $EF_{grid, CM, y}$ is 0.5596 tCO₂/MWh⁷

The total electricity generations, the electricity traction from the grid and the net electricity supplied to the grid are as follows:

2010				2011			
Emission Factor: 0.5596	Net Electricity Generation (MWh)*	Electricity traction from grid (MWh)	Net elect. Supplied to the grid (MWh)**	Emission Factor: 0.5596	Net Electricity Generation (MWh)*	Electricity traction from grid (MWh)	Net elect. Supplied to the grid (MWh)**
Jan	0	0	0	Jan	3,155.15	45.77	3,109.38
Feb	0	0	0	Feb	2,643.03	43.39	2,599.64
Mar	0	0	0	Mar	3,680.14	43.61	3,636.53
Apr	0	0	0	Apr	11,341.81	33.64	11,308.17
May	1,807.51	73.756	1,733.75	May	27,078.12	14.4	27,063.72
Jun	12,738.40	0.586	12,737.81	Jun	23,851.05	10.94	23,840.11
Jul	0	0	0.00	Jul	10,612.55	23.12	10,589.43
Aug	841.941	2.965	838.98	Aug	7,539.66	25.07	7,514.59
Sep	4,116.73	17.668	4,099.06	Sep	6,258.94	23.89	6,235.05
Oct	5,478.10	28.499	5,449.61	Oct	12,271.41	22.01	12,249.40
Nov	4,959.56	33.78	4,925.78	Nov	5,610.88	36.43	5,574.45
Dec	4,079.22	37.69	4,041.53	Dec	4,615.49	41.87	4,573.62
Total	34,021.46	194.94	33,826.52	Total	118,658.23	364.14	118,294.09

* Net electricity corresponds the measured electricity generation by the meters. This value does not include the internal consumption of the plant

** Net elct. Supplied to the grid = Net Electricity Generation - Electricity traction from grid

The figures have been obtained from the PMUM Records.

Table 2: The excel table displaying the PMUM records of Uzundere I HEPP from August 2010 to December 2011⁸

Based on the above electricity generation figures, baseline emission figure for 2010 and 2011 is calculated as follows:

$$BE_y = EG_y \times EF_{grid,y}$$

Where:

BE_y Baseline emissions (tCO₂e)

EG_y Annual electricity supplied by the project to the grid (MWh)

$EG_{baseline}$ Baseline electricity supplied to the grid in the case of modified or retrofit facilities (MWh). For new power plants this value is taken as zero.

$EF_{grid, CM, y}$ Baseline emission factor (tCO₂e/MWh)

Y Refers to a given year

As a result, based on the above electricity generation figures, baseline emission figure for 2010 and 2011 are calculated as follows:

$$\begin{aligned} BE_{2010} &= EG_{PJ,2010} \times EF_{grid,CM,2010} \\ &= 33,826.52 \text{ MWh} \times 0.559 \text{ tCO}_2/\text{MWh} \\ &= 18,929.32 \text{ tCO}_2 \end{aligned}$$

$$BE_{2011} = EG_{PJ,2011} \times EF_{grid,CM,2011}$$

⁸ The emission reductions are calculated based monthly PMUM (Market Financial Settlement Center) Records.

$$= 118,294.09 \text{ MWh} \times 0.559 \text{ tCO}_2/\text{MWh}$$

$$= \mathbf{66,197.37 \text{ tCO}_2}$$

During the monitoring period, all emission reductions were:

$$\mathbf{BE_{2010} + BE_{2011} = 18,929.32 + 66,197.37}$$

$$= \mathbf{85,126.69 \text{ tCO}_2}$$

4.2 Project Emissions

The Project emissions are calculated with the formula mentioned in ACM0002/Version11as:

$$PE_y = \frac{EF_{Res} \cdot TEG_y}{1000}$$

Where:

PE_y = Emission from reservoir expressed as tCO₂e/year

EF_{Res} = Is the default emission factor for emissions from reservoirs, and the default value as per EB23 is 90 Kg CO₂e /MWh

TEG_y = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh).

However, again according to ACM0002 / Version11, if the power density (PD) of the hydro power plant is above 10 W / m², PE_y is 0.

The power density of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity, in W/m²

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)⁹

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of

⁹ Electrical installed capacity of the project is used to calculate the power density of the project.

the project activity, when the reservoir is full (m^2)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation

of the project activity, when the reservoir is full (m^2). For new reservoirs, this value is 0.

Therefore,

$$Cap_{PJ} = 62,152,000 \text{ W}$$

$$Cap_{BL} = 0 \text{ (Justification: The project is a new hydro power plant)}$$

$$A_{PJ} = 7,275.54 \text{ m}^2 \text{ (Reservoir area at 1638 m altitude. Please see the technical drawing provided under Appendix 1)}$$

$$A_{BL} = 0 \text{ (Justification: The project is a new hydro power plant)}$$

Therefore;

$$PD = (62,152,000 - 0) / (7,275.54 - 0) = 8,542 \text{ W/ m}^2$$

Therefore;

Since the Power Density of the Project is greater than 10 W/m^2 PE_y is assumed to be 0 as suggested in ACM 0002 / Version 11

$$PE_{2011} = 0 \text{ tCO}_2/\text{year}$$

4.3 Leakage

Also, as suggested in ACM0002 / Version11, the leakage emissions are not considered and assumed as "0".

Therefore:

$$LE_{2011} = 0 \text{ tCO}_2/\text{year}$$

4.4 Summary of GHG Emission Reductions and Removals

Also, according to ACM0002/Version11 the emission reductions in year "y" should be calculated as the following formula

$$ER_y = BE_y - PE_y - LE_y$$

Where:

$$ER_y = \text{Emission reductions in year } y \text{ (t CO}_2\text{e/yr)}$$

$$BE_y = \text{Baseline emissions in year } y \text{ (t CO}_2\text{e /yr)}$$

PE_y = Project emissions in year y (t CO₂/yr)

LE_y = Leakage emissions in year y (t CO₂/yr)

Since PE_y and LE_y are assumed to be 0, emission reductions are equal to baseline emissions.

$$ER_{2010} = BE_{2010}$$

$$ER_{2011} = BE_{2011}$$

$$ER_{2010} = 18,929.32 \text{ tCO}_2$$

$$ER_{2011} = 66,197.37 \text{ tCO}_2$$

Please see the tables below for the emission reductions of the project for 2011:

Please see the table below for the emission reductions of the project between May 2010 and December 2011:

2010					2011				
Emission Factor: 0.5596	Net Electricity Generation (MWh)*	Electricity traction from grid (MWh)	Net elect. Supplied to the grid (MWh)**	Emission Reductions (Tons)	Emission Factor: 0.5596	Net Electricity Generation (MWh)*	Electricity traction from grid (MWh)	Net elect. Supplied to the grid (MWh)**	Emission Reductions (Tons)
Jan	0	0	0	0.00	Jan	3,155.15	45.77	3,109.38	1,740.01
Feb	0	0	0	0.00	Feb	2,643.03	43.39	2,599.64	1,454.76
Mar	0	0	0	0.00	Mar	3,680.14	43.61	3,636.53	2,035.00
Apr	0	0	0	0.00	Apr	11,341.81	33.64	11,308.17	6,328.05
May	1,807.51	73.756	1,733.75	970.21	May	27,078.12	14.4	27,063.72	15,144.86
Jun	12,738.40	0.586	12,737.81	7,128.08	Jun	23,851.05	10.94	23,840.11	13,340.93
Jul	0	0	0.00	0.00	Jul	10,612.55	23.12	10,589.43	5,925.85
Aug	841.941	2.965	838.98	469.49	Aug	7,539.66	25.07	7,514.59	4,205.16
Sep	4,116.73	17.668	4,099.06	2,293.83	Sep	6,258.94	23.89	6,235.05	3,489.13
Oct	5,478.10	28.499	5,449.61	3,049.60	Oct	12,271.41	22.01	12,249.40	6,854.76
Nov	4,959.56	33.78	4,925.78	2,756.47	Nov	5,610.88	36.43	5,574.45	3,119.46
Dec	4,079.22	37.69	4,041.53	2,261.64	Dec	4,615.49	41.87	4,573.62	2,559.40
Total	34,021.46	194.94	33,826.52	18,929.32	Total	118,658.23	364.14	118,294.09	66,197.37

* Net electricity corresponds the measured electricity generation by the meters. This value does not include the internal consumption of the plant

** Net elect. Supplied to the grid = Net Electricity Generation - Electricity traction from grid

The figures have been obtained from the PMUM Records.

Table 3.2: Emission Reductions by Uzundere I HEPP from May 2010 to December 2011

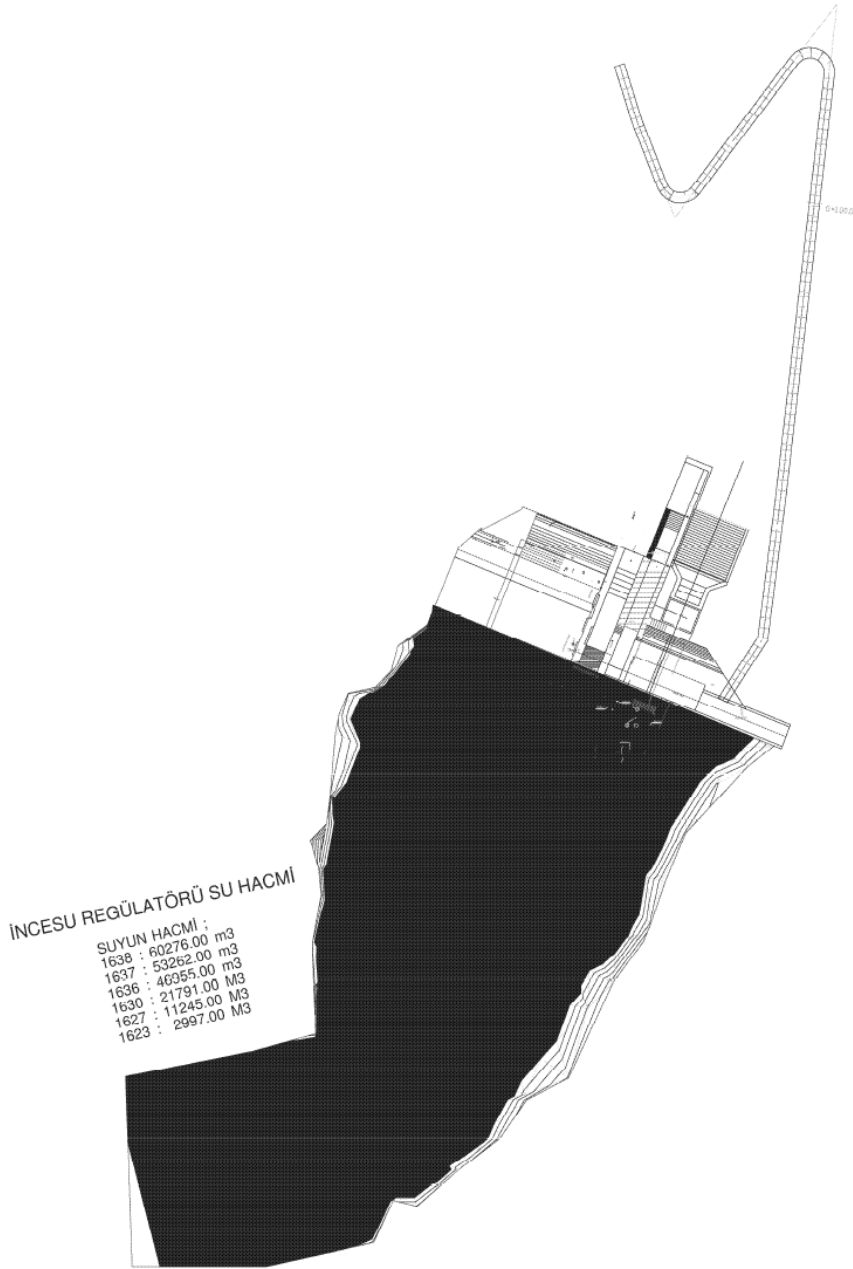
Total emission reductions realized as 85,126.69 tCO₂ for this monitoring period. When the estimated electricity generation figure of the power plant in the validated VCS PD (156,205 MWh/yr) is considered, the total emission reductions period should be approximately 145,000 tCO₂ for this 20 months. However, since the second turbine was not commissioned until November 2010, the electricity generations between May 2010 and November 2010 realized less than the figures in the FSR. On the other hand, when the full year for 2011 is considered, it is seen that the realized generation figure is 24% less than the estimated figure in the VCS PD. This also led to lower emission reduction. The reason why the realized figure is lower than the estimated figure is the less water availability. There is no problem with the efficiency of the power plant.

5 ADDITIONAL INFORMATION

N/A

APPENDIX-1

The following document explains that the reservoir area of Uzundere Hydroelectric power plant. The reservoir area has been calculated as 7,275.54 m² when the reservoir is at the fullest capacity. The drawing is obtained from the project plan prepared by EN-SU Muhendislik which is the company's consultant for etude and design of this project.



1638 KOTLU YÜZEY ALANI:7275.54m2